DSP · L08

Unilateral z-transform and mixed practice

Cheat sheet · Day 16 · Sun 8 Nov · pack D9

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

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UNILATERAL z-TRANSFORM (L08 s13-14)
X+(z) = sum_{n>=0} x(n) z^-n     (= X(z) if x causal; inverse ALWAYS causal, no ROC c
y(n-1) -> z^-1 Y + y(-1)
y(n-2) -> z^-2 Y + z^-1 y(-1) + y(-2)        <- don't drop the z^-1 y(-1) term
x(n-k) -> z^-k X   for input starting at n=0  (x(-1)=x(-2)=0)
x(n+1) -> zX - z x(0)

2nd order: A(z)Y + I(z) = B(z)X,  I(z) = a1 y(-1) + a2 y(-2) + a2 y(-1) z^-1
  Y_zs = B X / A = H X   (input+system poles)     Y_zi = -I / A  (system poles only)
  y = y_zi + y_zs

PAIRS  p^n u(n) <-> 1/(1-p z^-1)      (n+1)p^n u(n) <-> 1/(1-p z^-1)^2
       n p^n u(n) <-> p z^-1/(1-p z^-1)^2
RESIDUE A_i = (1-p_i z^-1)Y |_{z^-1 = 1/p_i}; double pole: C from squared factor,
        B from z^-1 = 0 (constant terms)
CHECKS  y(0) = Y(z^-1 = 0) = recursion value;  y(1) by recursion
        step steady state = H(1) (stable only)
L04 CHECK  modes + y_p, fit at y(0),y(1) from recursion (NOT at y(-1),y(-2))
           only y_zi may be fitted at n=-1,-2.  L04 constants != y_zi coefficients
KILL A MODE p: numerator of Y = 0 at z^-1 = 1/p  -> solve for unknown IC
ROC (bilateral): causal |z|>max pole; stable <=> ROC contains |z|=1;
                 causal & stable <=> all poles inside unit circle
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